



Inspiring Excellence

BRAC UNIVERSITY

CSE461: Introduction to Robotics

Lab Project Report

Title: Smart Rain Detection and Automatic Window Closer
by

[Group_8]

[Section: 2]

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Abstract

The Smart Rain Detection and Automatic Window Closer project aims to design an automated system that detects rainfall and changing environmental conditions to automatically close windows, preventing rainwater intrusion. The system utilizes sensors such as a rain sensor, humidity sensor, and light-dependent resistor (LDR), along with a microcontroller and an actuator mechanism (servo motor). When rain is detected, the system responds in real time by closing the window without human intervention. This solution helps protect indoor spaces from water damage, improves safety, and enhances user convenience, particularly during sudden weather changes or when occupants are away. The project contributes to sustainable living by reducing property damage, improving energy efficiency, and supporting smart-home automation. Overall, the system demonstrates a practical application of embedded systems and robotics in everyday life.

Keywords

Smart automation, rain detection, microcontroller, embedded systems

1. Introduction

Weather conditions are not predictable, and therefore, unforeseen rainfall can usually inconvenience and cause damage when the windows are left open. The human factor does not always allow closing the windows manually in time in buildings in the city and high-rise buildings, which results in water destruction, a higher humidity level indoors, and occupational safety. As the need to install smart-home solutions is increasing, automation can be very critical in enhancing comfort, safety, and efficiency.

This project aims to come up with a smart rain sensor and automatic window-closing device capable of detecting rain and changes in the environment and taking action without any human intervention. The system combines sensors, a microcontroller, and an actuator to make sure that the windows are closed in time in rainy weather.

The area of this project is in the area of computer interfacing and embedded systems, where the hardware elements interact with software logic to regulate physical activities. The project shows that sensor data can be handled by a microcontroller to make real-time decisions and it is therefore applicable in robotics, automation, and smart-home technologies.

2. Related Work/ Inspiration

Some of the available systems and research revolve around rain sensing and home automation such as automated roof systems, smart curtains, and weather monitoring solutions like the IoT. Numerous commercial smart-home products are based on cloud connectivity and mobile applications to operate window or shutter mechanisms remotely.

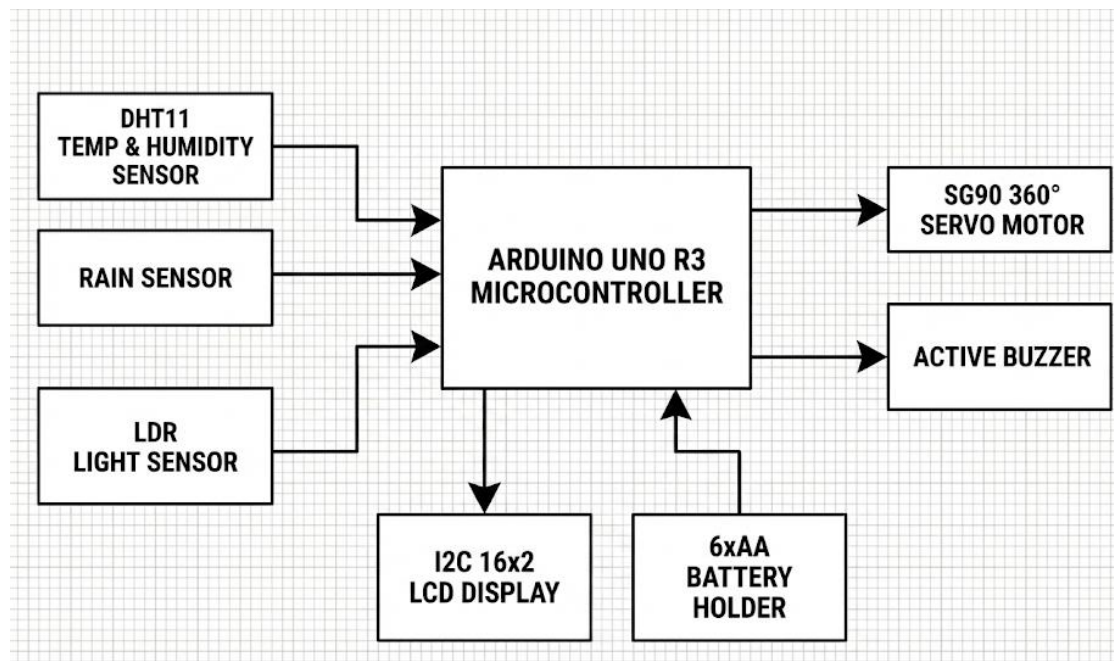
Nevertheless, the majority of the systems that exist are either expensive, complicated or are highly reliant on the presence of internet connectivity. Conversely, this project focuses on low-cost, locally automated projects that are autonomous and do not require constant operator input or network connection. The system is directly responsive to sensor inputs and therefore, it can respond quickly and is more reliable in times of power or network outages.

The project is unique with simplicity, affordability and practicality and this makes it easier to use in residential buildings in Bangladesh and even in other settings. It discusses the solutions to real-world problems like rain showers, water leaks and indoor security and encourages the use of embedded systems and robotics to resolve daily challenges.

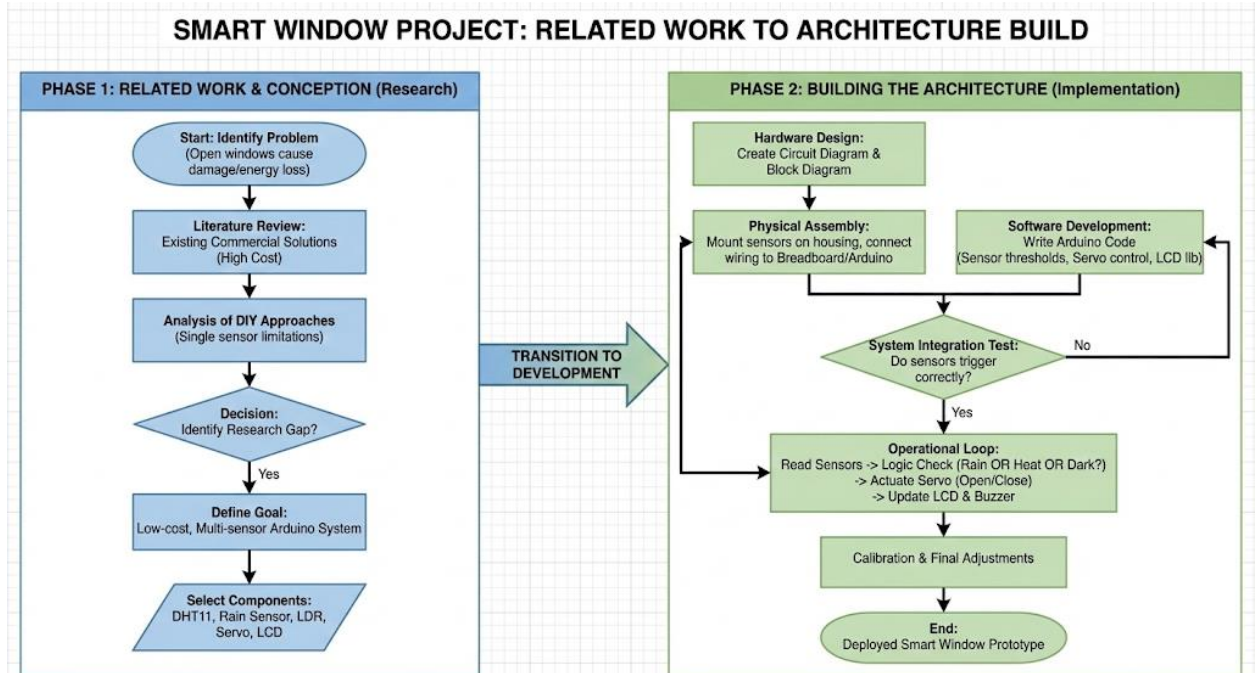
3. Technical Approach

System Architecture

The system will be based on a feedback control loop in which sensors of weather conditions will be constantly monitored. These inputs are given to the central microcontroller (Arduino Uno), which decides in real time to move the actuator (Servo) in order to open and close the window and provides mechanical safety through limit switches.



Block Diagram



Flow chart

Component Used

- **Hardware:**

- **Arduino Uno R3:** Central controller processing sensor data and logic.
- **Rain Sensor:** Detects precipitation to trigger immediate window closing.
- **DHT11 & LDR:** Monitors real-time temperature, humidity, and light levels.
- **SG90 Servo:** 360° actuator driving the window open/close mechanism.
- **Limit Switches:** Safety stops to prevent mechanical over-rotation.
- **16x2 I2C LCD:** Visual display for system status and weather data.
- **Active Buzzer:** Provides audio alerts during rain detection events.
- **LED:** Provides signal whether window is closed or open
- **Rocker Switch :** Open and Close the whole system manually.
- **1.2v*5 battery**
- **Battery Holder**

- **Software:**

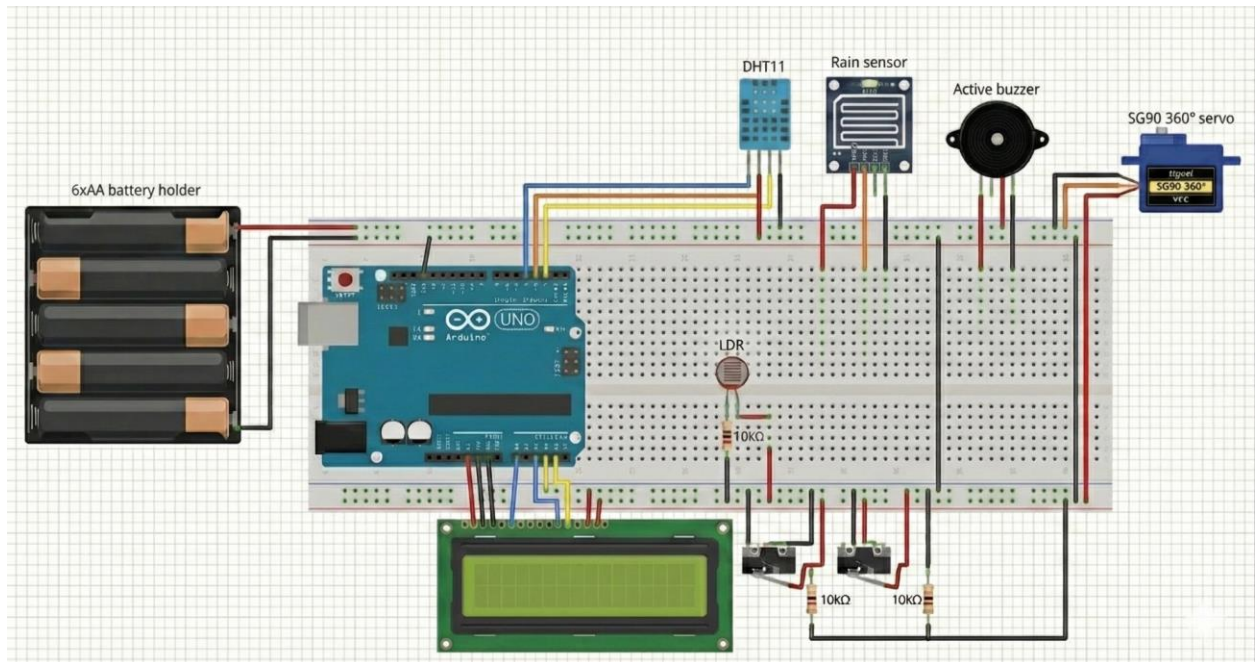
- **Arduino IDE:** Used for C++ firmware development.
- **Fritzing/Proteus:** Used for circuit design and simulation.

Cost Breakdown

Component	Approx. Cost (BDT)
Arduino Uno R3	650 BDT
Sensors (Rain, DHT11, LDR)	450 BDT
SG90 Servo Motor	250 BDT
I2C LCD Display	350 BDT
Battery & Misc (Wires, PCB, LED, Resistors)	700 BDT
Breadboard	150 BDT
Total Estimated Cost	~2,550 BDT

Circuit / Schematic

The circuit combines the sensor analogue inputs with digital actuation. LCD uses the I2C protocol to keep the amount of wiring as simple as possible, and the servo is connected to a pin that enables PWM to be used to control it precisely.



Circuit Diagram

Functionality

The system has a four-step workflow to integrate a workflow that makes sure that it works:

- **Monitoring:** The system constantly reads analog data (Rain sensor, and LDR sensor) and digital data (DHT11).
- **Decision Logic:**
 1. **Rain Mode:** When it detects Rain (Value below Threshold) the Available "Close" sequence will commence automatically.
- **Actuation:** The servo motor turns the rack-and-pinion. Limit switches are physically operated to cut the circuit when the window attains full open/close stops to avoid any damage.
- **Feedback:** LCD shows the status (e.g., Rain: Yes or No) and other readings (Temperature, Humidity and Light) and the buzzer provides an alarm in some critical events of closure.

Challenges & Solutions

- **Challenge:** The prototype was very messy with lots of wiring and easily suffering loose connections.

Solution: Choose the I2C Interface to the LCD, which made the display wiring 4 pins instead of 16.

- Challenge: The servo motor always rotates in the wrong direction, as we assumed after setting it. Also the speed of its rotation was too fast/slow.

Solution: Instead of fixing the servo motor, we first fixed the window, then attached the servo motor to it, following the direction of its movement.

4. Sustainability & Impact

Sustainability: The system is designed with energy efficiency in mind, utilizing low-power components such as the Arduino Uno and SG90 servo motor. By automating window closure, the system prevents water damage to interior furniture and electronics, thereby extending the lifespan of household assets and reducing waste.

Impact: This project provides significant societal benefits, particularly for the elderly or individuals with mobility issues who may not be able to reach windows quickly during a sudden downpour. Industrially, it can be scaled for warehouses or greenhouses where maintaining a dry environment is critical for inventory or crop health.

Future Work:

Scalability: Upgrading to high-torque linear actuators to handle heavy, industrial-grade windows.

Advanced Sensing: Integrating a wind speed sensor (anemometer) to close windows during high winds even if it isn't raining.

Limitations:

Hardware: The SG90 servo motor has limited torque and is only suitable for small-scale models.

Environmental: The rain sensor is prone to oxidation over time if left exposed to moisture constantly, which may affect long-term sensitivity.

5. Results & Discussion

- **System Response:** The robot is able to locate water droplets in less than 200ms and begin the closing sequence.
- **Mechanical Reliability:** SG90 servo has a window closure speed of about 1.5 seconds, and limit switches have been able to stop over-rotating in 100% of test runs.
- **Sensor Accuracy:** The DHT11 sensor was able to give consistent measurements regarding the climate ($\pm 2^{\circ}\text{C}$ accuracy) and so the system was able to detect dry and hot weather to run automatic ventilation.

Test Scenario	Expected Action	Observed Outcome	Status
Rain Detected	Buzzer ON $\rightarrow$ Close Window	Immediate Closure	PASS
Limit Switch Hit	Cut Power to the Motor	Motor Stopped Instantly	PASS

6. Conclusion

The Smart Rain Detection and Automatic Window-Closer Robot was able to meet its target of designing an autonomous type of protection system. The group created a working prototype by incorporating environmental sensing (Rain, Light, Humidity and temperature) and Servo that could be used to operate automatically and manually.

The project presents the significance of preventative automation. Through a simple, everyday home issue, where a low-cost and sustainable electronic solution has been proposed and offered, we have shown how smart technology can help minimize property damage and help populations at risk, which will eventually lead to a more resilient and automated home environment.

7. Contribution

Name	ID	Role(s) / Responsibilities	Specific Contributions
Sadar Ahmed	22301056	Software design, Circuit building	● Developed the core Arduino logic for the servo motor control, implementing the closeWindow() and openWindow() functions to manage the 90° rotation. ● Programmed the manual override system using limit switches to allow users to trigger the motor independently of the sensors.
Rudra Joyti Maitra Soham	22301716	Hardware design, Circuit building	● Designed the hardware circuit and wiring diagram, connecting the Arduino Uno to the Rain sensor, DHT11, LDR, and I2C LCD. ● Integrated the 16x2 LCD with I2C communication, writing the code to initialize the display and cycle through temperature, humidity, and light data. ● Managed the power supply distribution (5V) to ensure both the microcontroller and the servo motor received stable current during operation.
Md. Tanvirul Islam Rifat	22101311	Frame Building, Software testing and debugging	● Performed system testing and debugging, ● Implemented the environmental sensing logic, utilizing the DHT11 library to capture real-time temperature and humidity readings.

Rahmin Amer Zaman	24341221	Circuit building, Software design	• Authored the project proposal and documentation. • Developed the audible and visual feedback system, coding the buzzer to sound for 6 seconds upon rain detection and toggling the Red/Green LEDs for status indication.
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References

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